Check for updates

Jay Patrikar
Robotics Institute,
Carnegie Mellon Institute,
Pittsburgh, PA 15213
e-mail: jaypat@cmu.edu

Venkata Ramana Makkapati
School of Aerospace Engineering,
Georgia Institute of Technology,
Atlanta, GA 30332
e-mail: mvramana@gatech.edu

Anay Pattanaik
Coordinated Science Lab,
University of Illinois at Urbana-Champaign,
Urbana, IL 61801
e-mail: anayp2@illinois.edu

Hardik Parwana
Department of Mechanical
Engineering and Science,
Kyoto University,
Kyoto 606-8501, Japan
e-mail: parwana.hardik.22r@st.kyoto-u.ac.jp

Mangal Kothari[2]
Department of Aerospace Engineering,
Indian Institute of Technology,
Kanpur, Uttar Pradesh 208016, India
e-mail: mangal@iitk.ac.in

Nested Saturation Based Guidance Law for Unmanned Aerial Vehicles[1]

In this paper, using the theory of input saturation, a novel path following guidance law for fixed-wing unmanned aerial vehicles (UAVs) is developed. The proposed guidance law is adapted from a pursuit plus line-of-sight guidance law. Furthermore, it employs inertial speed for computing the acceleration commands which adds an adaptive capability of accommodating vehicle speed changes due to external disturbances such as wind. The guidance law is initially developed for two-dimensional (2D) environments which enables vehicles to follow straight lines, circles, and ellipses in planar spaces. Lyapunov theory is used to establish its stability properties, followed by a comparative study with existing algorithms, proposed for 2D environments, to establish its efficacy. The guidance law is then extended for the case of three-dimensional (3D) environments, and appropriate simulation studies are performed. Finally, real-world flight tests for 2D as well as 3D cases are presented, establishing the applicability of the proposed law on UAVs.
[DOI: 10.1115/1.4043107]

Downloaded from https://asmedigitalcollection.asme.org/dynamicsystems/article-pdf/141/7/071008/6415992/ds_141_07_071008.pdf by guest on 05 July 2020

1 Introduction

By the year 2020, an estimate of about 1 million unmanned aerial vehicles (UAVs) is expected to roam the sky carrying out various tasks including traffic monitoring, package delivery, crop-spraying, surveying, and disaster research and management.[3] Among several different kinds of UAVs (including single and multirotor), fixed-wing UAVs have proven to be superior in terms of their long-range and endurance capabilities. They provide effective solutions in the aforementioned applications, accelerating their deployment [1–5]. Almost every other UAV-related solution requires that they follow a predetermined path in three-dimensional (3D) environments, and as a result, path following and trajectory tracking algorithms play a crucial role in the development of autonomous UAVs. The path following problem is concerned about spatial tracking, while the trajectory tracking handles an added temporal constraint. In this paper, we formulate and solve the path following problem in 3D environments.

There is a plethora of linear and nonlinear guidance laws (NLGL) for path following available in the literature, especially for two-dimensional (2D) environments. Path following techniques employing proportional-integral-derivative control and linear quadratic regulator theory fall into the realm of linear guidance laws [6–8]. The nonlinear guidance laws primarily employ techniques based on (i) error regulation, (ii) vector fields, and (iii) virtual targets [9]. Guidance laws that regulate error variables, including cross-track and/or heading errors, using nonlinear control theory though guarantee stability and performance, they are model-dependent and pose certain challenges for real-time implementation [10–14]. Alternatively, vector fields can be used to designate the desired velocity at each and every point that will allow the vehicle to converge to the desired path [15,16]. The vector field based guidance laws have been developed for specific curves (such as straight lines and circles), and they cannot be easily extended to any general curve. Finally, path following techniques inspired from the missile guidance theory, such as pure pursuit [17,18], line-of-sight [19–22], and proportional navigation [23], which are geometric, employ virtual targets on the desired path [24,25]. Additionally, path shaping methods [8], dynamic inversion [26], neural networks [27], and differential flatness [28,29] have been used to develop guidance laws. A comparative study on the existing guidance laws for 2D environments, delineating the pros and cons of each technique, can be found in Ref. [30].

The literature on path following for 3D environments is rather limited. Cho et al. developed a simple yet efficient 3D path-following guidance law that enables exact tracking of a space curve satisfying certain smoothness conditions [9]. Breivik and Fossen made an attempt to provide an algorithm for path following in a general n-dimensional space [25]. Pelizer et al. [31] presented a comparative study (restricted to straight paths) on the performance of selected guidance laws in 3D environments [31]. Also, a majority of research do not consider the effect of input

[1]The work has been carried out at Intelligent Guidance & Control Laboratory, Indian Institute of Technology Kanpur (IIT Kanpur).
[2]Corresponding author.
[3]https://www.nasa.gov/ames/utm

Contributed by the Dynamic Systems Division of ASME for publication in the Journal of Dynamic Systems, Measurement, and Control. Manuscript received March 25, 2018; final manuscript received February 17, 2019; published online April 3, 2019. Assoc. Editor: Vladimir Vantsevich.

Copyright © 2019 by ASME

saturation which results in performance degradation of the guidance law in real-world applications. A reduction in the effective area of convergence due to input saturation was demonstrated by Park et al. using Lyapunov stability analysis [24]. Zhao et al. presented two approaches for path following under control constraints: one uses a tracking equation defined in Frenet–Serret frame coupled with a stabilizing guidance law with control damping and the other uses the nested saturation theory [32]. However, the work is limited to 2D environments and does not involve any experiments on hardware. An attempt has also been made by Beard et al. to address the problem of input saturation using the nested saturation techniques but their approach constrained the roll and the flight path angles (as opposed to lateral and normal accelerations) and calculated the desired roll and flight path angles for path following in the longitudinal and lateral planes [14]. The presented flight test results are limited to constant altitude maneuvers, and the stability analysis has theoretical inconsistencies. Our approach differs with its uniform treatment of the decoupled cases by calculating the desired accelerations instead of the angles, and we also provide details on the subsequent use of more established lower level controllers to calculate the actuator commands.

The integrated guidance and control loop architecture is one of the two methodologies that is used to achieve path following [11,33]. The approach fuses the guidance law and the controller, and further requires accurate model data, making it hard to implement on real-time flight systems. The alternative approach, followed in this paper, exploits the time-scale separation between translational and rotational dynamics to separate the guidance law (outer loop) that generates the acceleration commands, from the controller (inner loop) and that tracks the generated commands. An added advantage with the latter is that the existing autopilot(s) can employ any given path following guidance law as long as the inner loop meets certain requirements which are coupled with the outer loop performance specifications.

The main contributions of this work can be summarized as follows. Owing to the fact that the longitudinal and the lateral dynamics of a fixed-wing UAV are decoupled, the 3D path following problem is solved by first projecting the desired path onto two orthogonal planes in a local frame, essentially reducing it to a 2D path following problem. In this regard, a novel guidance law based on pursuit plus line-of-sight guidance for 2D environments that explicitly accounts for input saturation is presented in this paper. Nested saturation theory is employed to address the problem of input saturation, and using LaSalle's theory, a rigorous convergence proof is presented to establish guaranteed asymptotic stability. The proposed approach draws its motivation from the earlier work by Teel who gave a method for constructing a control law that guarantees global asymptotic stability of linear systems with bounded controls [34]. To the best of authors' knowledge, there does not exist a complete proof for path following solutions employing nested saturation techniques. In this paper, the analysis is extended to nonlinear systems. Furthermore, inertial speed is incorporated in computing the acceleration commands to account for the effect of variable wind and render the scheme adaptive to external disturbances. We demonstrate a superior performance of the proposed law over adaptive optimal guidance law (AOGL) [8], pursuit plus line-of-sight law (PLOS) [35], vector field path following (VFPF) [15], and NLGL [24]. As the convergence is guaranteed in 2D environments, it can be argued that the guidance law, when employed on a fixed-wing UAV with decoupled dynamics, ensures convergence for paths in 3D environments. Numerical simulations and extensive flight tests on a fixed-wing platform, conducted at Flight Laboratory, Indian Institute of Technology Kanpur, for 3D paths show the efficacy of the proposed approach. Preliminary results of this work were presented in Ref. [36] addressing the problem with input constraints in 2D cases. The current work extends the solution to 3D environments, along with experimental results on a real-time system (fixed-wing UAV), which are not a part of our previous work.

The rest of the paper is organized as follows. Section 2 contains the problem formulation and the solution approach. Section 3 presents the proposed guidance law along with the main result that establishes its convergence properties in 2D cases. Section 4 demonstrates the performance of the proposed law with extensive simulations and a comparative study. Section 5 discusses the flight test results, and Sec. 6 concludes the paper. The complete proof for the main result can be found in the Appendix.

2 Problem Formulation

Consider a UAV flying in the three-dimensional Euclidean space as shown in Fig. 1. The kinematics of the UAV are given by

$$\dot{x} = v\cos\gamma\cos\chi \tag{1a}$$

$$\dot{y} = v\cos\gamma\sin\chi \tag{1b}$$

$$\dot{z} = v\sin\gamma \tag{1c}$$

$$\dot{\chi} = \frac{u_h}{v\cos\gamma} \tag{1d}$$

$$\dot{\gamma} = \frac{u_v}{v} \tag{1e}$$

where $(x, y, z) \in \mathbb{R}^3$ denotes its position in the inertial frame, defined by the coordinate system XYZ (see Fig. 1) and X–Y is the horizontal plane which is parallel to the ground. Further, v denotes the inertial speed, γ is the elevation angle, and χ is the azimuth angle. The velocity vector of the UAV $\mathbf{V}$ can be defined using the inertial speed ($v = |\mathbf{V}|$), elevation and azimuth angles (γ and χ). Finally, u_h and u_v are the commanded lateral accelerations which are applied independently and are limited by $|u_h| \leq u_{h_{\max}}$ and $|u_v| \leq u_{v_{\max}}$. It is assumed that at every instant of time, there is full-state feedback, includes (x, y, z) and the angles γ, χ, and the inertial speed v along with the desired path is available to us. The objective of the path following problem is to find the commanded accelerations that guarantee the convergence of the UAV's path to its desired one. To this end, we first define a noninertial local coordinate system $X_\ell Y_\ell Z_\ell$ with the UAV's position as its origin, and X_ℓ parallel to the velocity vector. Y_ℓ and Z_ℓ are, respectively, parallel to the directions in which the commanded accelerations u_h and u_v are applied. The rotation matrix which can be used to transform from the inertial frame to the local frame is given by

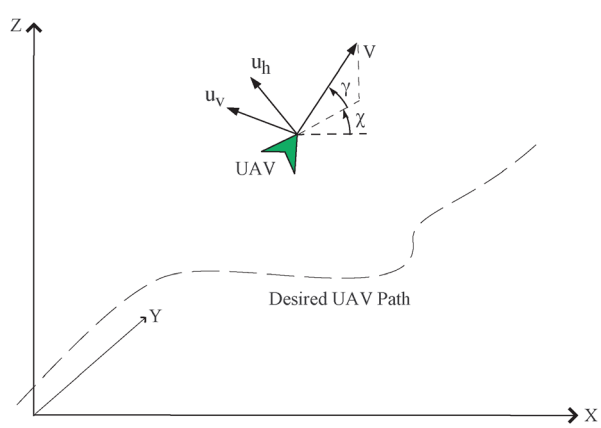

Fig. 1 UAV in a 3D environment

Downloaded from https://asmedigitalcollection.asme.org/dynamicsystems/article-pdf/141/7/071008/6415992/ds_141_07_071008.pdf by guest on 05 July 2020

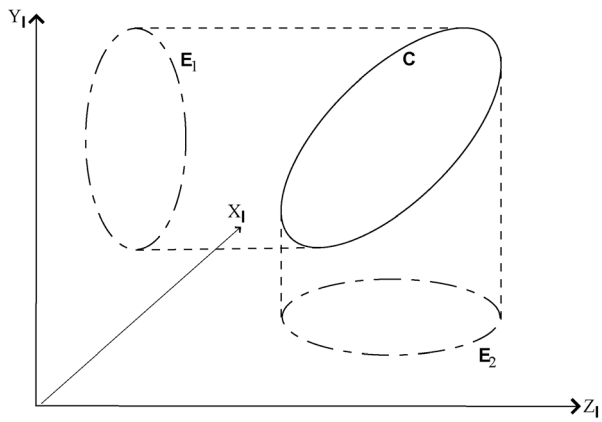

Fig. 2 Projections of a circle in 3D: $\mathcal{C}$ represents the circle, $\mathcal{E}_1$ and $\mathcal{E}_2$ are the ellipses on the projected planes

$$R_i^\ell = \begin{bmatrix} \cos\gamma\cos\chi & \cos\gamma\sin\chi & \sin\gamma \\ -\sin\chi & \cos\chi & 0 \\ -\sin\gamma\cos\chi & -\sin\gamma\sin\chi & \cos\gamma \end{bmatrix} \tag{2}$$

Given a feasible path, it is now possible to take its projection on the $X_\ell - Y_\ell$ (local horizontal) and $X_\ell - Z_\ell$ (local vertical) planes, in local frame. Since the commanded accelerations are decoupled, the projections are then used to solve the corresponding 2D path following problems by generating commanded accelerations that affect the dynamics only in their respective planes. Thus, given a 3D path following problem, we can solve two reduced order 2D path following problems simultaneously to arrive at a feasible solution. To this end, the cross-track and the heading errors are obtained for the projected paths in their respective planes, as discussed in Secs. 2.1–2.3. The convergence in the individual planes will ultimately guarantee the convergence with the desired path in 3D. The projection of a straight line path in 3D on a plane is another straight line, while the projection of a circle in 3D is an ellipse as shown in Fig. 2.

Consequently, the 2D path following problem in the Euclidean plane (X–Y) is considered with the following planar kinematics:

$$\dot{x} = v\cos\psi \tag{3a}$$

$$\dot{y} = v\sin\psi \tag{3b}$$

$$\dot{\psi} = \frac{u}{v} \tag{3c}$$

where ψ is the heading angle measured from the X-axis, as shown in Fig. 3, and it is used for generality in place of γ and χ. u is the commanded lateral acceleration which is limited by $|u| < u_{\max}$. Figure 3(a) represents the case when the current heading is "consistent" with the desired heading, and Fig. 3(b) represents when it is otherwise. The instantaneous coordinate system P_xP_y is defined with respect to the point on the desired path that is closest to the UAV. As shown in Fig. 3, the axis P_x is along the tangent to the desired path at the closest point, and P_y is perpendicular to P_x and along the line joining the closest point and the UAV. The cross-track error d is the shortest distance to the desired path. The heading error is defined as $\tilde{\psi} = (\psi - \psi_r)$, where ψ_r represents the required heading angle at the closest point on the desired path. Formally, the 2D path following problem is then to compute the required lateral acceleration u, satisfying the conditions

(1) $\lim_{t\to\infty} d = 0$,
(2) $\lim_{t\to\infty} \tilde{\psi} = 0$,
(3) $|u| < u_{\max}$,

subject to the kinematics in Eq. (3).

The rate of change of the cross-track error $\dot{d}$ and the acceleration of the cross-track error $\ddot{d}$ for the case shown in Fig. 3(a) can be obtained as

$$\dot{d} = v\sin(\psi - \psi_r) = v\sin\tilde{\psi} \tag{4a}$$

$$\ddot{d} = u\cos(\psi - \psi_r) = u\cos\tilde{\psi} \tag{4b}$$

and for the case in Fig. 3(b), they are

$$\dot{d} = v\sin(\psi - (\psi_r - \pi)) = -v\sin\tilde{\psi} \tag{5a}$$

$$\ddot{d} = u\cos(\psi - (\psi_r - \pi)) = -u\cos\tilde{\psi} \tag{5b}$$

The efficacy of a path following guidance law is established using two standard desired paths, the straight line (way-point guidance) and the circle (loiter guidance). As the projection of a

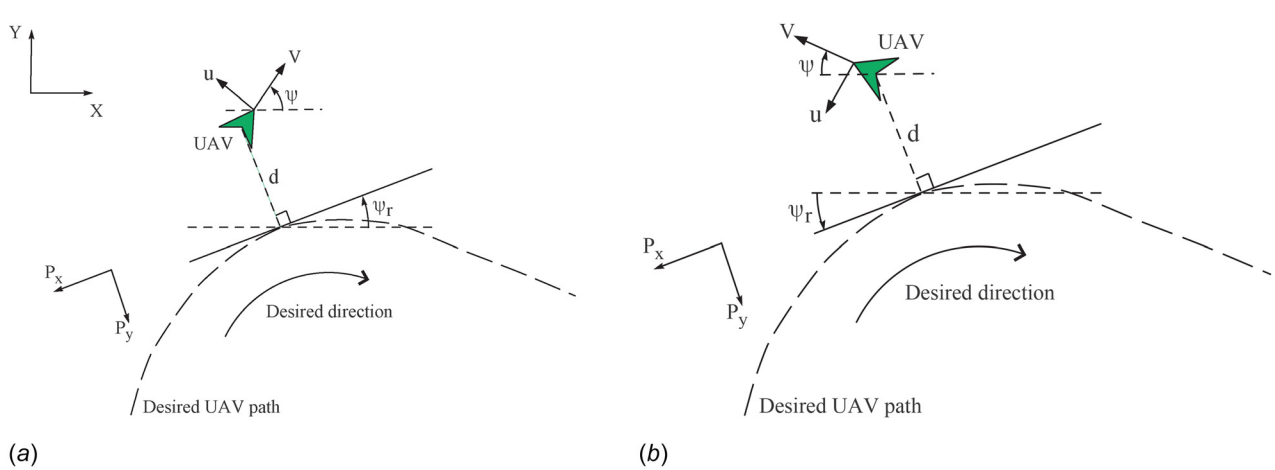

Fig. 3 Schematics of the two possible path following scenarios: (*a*) current heading is consistent with the desired one and (*b*) current heading is inconsistent with the desired one

Downloaded from https://asmedigitalcollection.asme.org/dynamicsystems/article-pdf/141/7/071008/6415992/ds_141_07_071008.pdf by guest on 05 July 2020

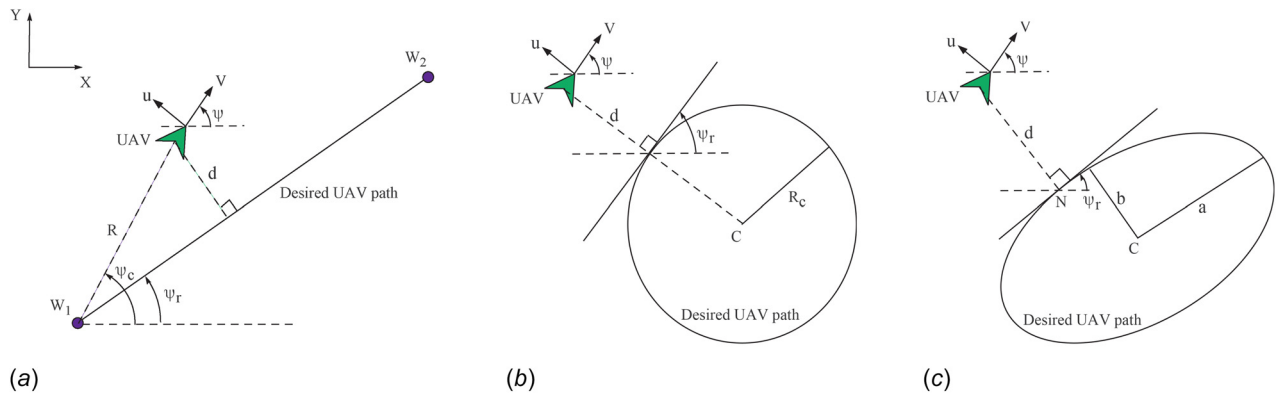

Fig. 4 Schematics of the 2D path following geometries: (*a*) straight line, (*b*) circle, and (*c*) ellipse

circle in 3D onto a plane is an ellipse, we also consider an elliptical path. The heading and the cross-track errors for the specific cases of straight line, circle, and ellipse (as the desired paths) are presented in the Secs. 2.1–2.3.

2.1 Straight Line Following. Consider a straight line path as shown in Fig. 4(*a*), and it can be observed that the heading is consistent with the desired direction. The UAV is commanded to follow the line joining the waypoints W_1 and W_2, and ψ_r is its slope. Let R be the displacement of the UAV from the waypoint W_1 and ψ_c be the slope of the line joining the UAV and W_1, measured from the X-axis. Then, the cross-track and heading errors can be obtained as

$$d = R\sin(\psi_c - \psi_r) \tag{6a}$$

$$\tilde{\psi} = \psi - \psi_r \tag{6b}$$

A similar set of expressions can be obtained when the heading is inconsistent with the line joining the waypoints W1 and W2, which are dropped for brevity, as followed in Secs. 2.2 and 2.3.

2.2 Circle Path Following. Consider a circular path centered at C (x_c, y_c) with radius R_c, as shown in Fig. 4(*b*). Assuming the heading ψ is consistent with the desired heading, the cross-track and heading errors can be obtained as

$$d = \sqrt{(x - x_c)^2 + (y - y_c)^2} - R_c \tag{7a}$$

$$\tilde{\psi} = \psi - \psi_r \tag{7b}$$

where

$$\psi_r = \tan^{-1}\left\{\frac{y_c - y}{x_c - x}\right\} \pm \frac{\pi}{2} \tag{8}$$

As per the mission requirement, the UAV may have to move either clockwise or anticlockwise, which determines the sign in front of $\pi/2$ in the above equation.

2.3 Ellipse Path Following. Consider an ellipse centered at C (x_c, y_c) with a and b as the lengths of the semimajor and the semiminor axes, respectively, as shown in Fig. 4(*c*). In order to find the cross-track error, the fact that the line joining the UAV and the point on the ellipse is perpendicular to the tangent at the closest point (to the UAV on the ellipse) is employed. Mathematically, given the current location of the UAV (x, y), the closest point on ellipse $N(x_p, y_p)$ satisfies the expression

$$x_p b^2 (y_p - y) = y_p a^2 (x_p - x) \tag{9}$$

subject to

$$\frac{x_p^2}{a^2} + \frac{y_p^2}{b^2} = 1 \tag{10}$$

It can be observed that solving for (x_p, y_p) using Eqs. (9) and (10) leads to a polynomial that can have more than one real root, each associated with a point on the ellipse. Among the set of obtained points, the one that is closest to the UAV is chosen. The desired heading angle ψ_r is the slope of the tangent to the ellipse at the closest point N. Finally, the errors can be obtained as

$$d = \sqrt{(x_u - x_p)^2 + (y_u - y_p)^2} \tag{11a}$$

$$\tilde{\psi} = \psi - \psi_r \tag{11b}$$

where

$$\psi_r = -\tan^{-1}\left\{\frac{x_p b^2}{y_p a^2}\right\} \tag{12}$$

3 Proposed Guidance Law

In this section, we develop a path following guidance law that explicitly accounts for control saturation, and it is based on pursuit PLOS guidance law. The proposed guidance law is motivated from the work in Ref. [34] that addresses the problem of stabilizing linear systems with bounded controls.

In order to satisfy the input constraint, a guidance law has to maintain $|u| \le u_{\max}$. An intuitive approach is to use a saturation function that limits the maximum commanded acceleration. However, the convergence of the path following algorithm cannot be guaranteed with the saturation function. The theory of nested saturation can be employed to mitigate this situation which requires the system dynamics to form of a chain of integrators [34]. At this point, the reader can refer to Definition 1 and Theorem 2.1 in Ref. [34], for the definition of a *linear saturation* (L, M) and for the result in the case of linear system consisting of multiple integrators, respectively. Throughout the paper, we refer to the specific case of $L = M$.

First, from Eqs. (4*a*) and (5*a*), it can be observed that the heading error is proportional to the rate of change of cross-track error: $\tilde{\psi} = \pm\sin^{-1}(\dot{d}/V)$. Consequently, instead of heading error (as suggested in PLOS), we aim to drive the rate of change of

Downloaded from https://asmedigitalcollection.asme.org/dynamicsystems/article-pdf/141/7/071008/6415992/ds_141_07_071008.pdf by guest on 05 July 2020

cross-track error $\dot{d}$ to zero. The modified control objectives for the nested saturation guidance law (NSGL) then become

(1) $\lim_{t\to\infty} d = 0$
(2) $\lim_{t\to\infty} \dot{d} = 0$

Second, from Eqs. (4*b*) and (5*b*), it can be observed that the behavior of the system with states $(d, \dot{d})$ is similar to that of a double integrator system; however, there is a nonlinearity associated with the input in the form of $\cos\tilde{\psi}$. One can use nonlinear techniques to synthesis control though it is not always possible to show convergence under saturation. As the nested saturation technique is suitable for cascaded integrators with bounded inputs, the following path guidance law is proposed:

$$u = -\sigma_{M_1}\left(\frac{k_1\dot{d} + \sigma_{M_2}(\zeta)}{\cos\tilde{\psi}}\right) \tag{13}$$

where $\zeta = k_2(k_1\dot{d} + d)$ and

$$\sigma_{M_i}(r) = \begin{cases} M_i, & r \geq M_i, \\ -M_i, & r \leq -M_i, \\ r, & \text{otherwise} \end{cases} \tag{14}$$

Here, M_1 is the limit on acceleration, i.e., $M_1 = u_{\max}$, k_1, k_2, and M_2 are design parameters which will be selected through Lyapunov analysis. Note that when both the nested functions σ_{M_1} and σ_{M_2} are unsaturated, the proposed law is similar to PLOS where $\ddot{d} = (k_1 + k_1k_2)\dot{d} + k_2 d$, which is proportional to a weighted average of cross-track error and rate of change of cross-track error. The main result of this paper is presented below.

THEOREM 3.1. *Given the error dynamics $\ddot{d} = \pm u\cos\tilde{\psi}$, and the constraint $u \leq M_1$, the system is Lyapunov stable for the control law in Eq. (13) when the gains k_1, k_2, and M_2 satisfy the conditions*

(1) $\dfrac{M_2}{k_1} < v_m$

(2) $$\frac{\dfrac{M_2}{k_1}}{\sigma_{M1}\left(\dfrac{M_2 + \sigma_{M2}(k_1M_2)}{\sqrt{1 - \dfrac{M_2{}^2}{v_m^2 k_1{}^2}}}\right)\sqrt{1 - \dfrac{M_2{}^2}{v_m^2 k_1{}^2}}} < \frac{k_1 + k_1k_2}{k_2}$$

where v_m is the maximum possible inertial speed of the UAV.

Sketch of the Proof. We divide the state space $\mathcal{S}$ into two regions, $\mathcal{S}_i \triangleq \{(d,\dot{d}) : |\dot{d}| \leq (M_2/k_1)\}$ and $\mathcal{S}_o \triangleq \{(d,\dot{d}) : (M_2/k_1) < |\dot{d}| \leq v_m\}$. First, it is shown that the trajectory will always enter the region $\mathcal{S}_i$. Next, we divide $\mathcal{S}_i$ into different sets based on possible directions of the error trajectory. It is then shown that under the above conditions on the gains k_1, k_2, and M_2, the stationary point is unique, and it is the largest invariant set in $\mathcal{S}_i$. Then, the convergence is established using LaSalle's theorem.

The complete proof is presented in the Appendix.

4 Simulation Results

In this section, we present the simulation results demonstrating 2D as well as 3D path following capabilities of the proposed guidance law. The 2D simulation results on straight line and circular paths include comparison with the existing 2D guidance laws NLGL, AOGL, VFPF, and PLOS to establish the superior performance of the proposed law. Finally, straight line and circular geometries in 3D are considered along with the special case of an elliptical path in 2D.

4.1 Two-Dimensional Straight Line. The UAV is commanded to follow a straight line with waypoints W_1(0, 0)m to W_2(300, 300)m (black solid line Fig. 5(*a*)), and its cruise speed is 15 m/s. A variable wind component of 4.5±0.4 m/s at an angle of 135 deg with respect to the *X*-axis is added to the simulation to model the external wind disturbances. The maximum lateral acceleration constraint is chosen to be $u_{\max} = 5$ m/s^2. The start point of the UAV is set at (20,0)m (triangle in Fig. 5(*a*)) with the initial velocity vector parallel to the *X*-axis. The gains are set to $k_1 = 0.8$, $k_2 = 0.7$, and $M_2 = 7$. The simulation results for NSGL are presented in Fig. 5, while comparing it with the existing guidance laws. Figure 5(*a*) shows the desired versus actual paths. The position error for NSGL converges to 0.7 m which is the acceptable range for most practical applications. As can be seen from Fig. 5(*a*), in terms of the commanded rate of lateral accelerations, NSGL definitely performs better than VFPF and AOGL. While AOGL has singularities in demanded accelerations, VFPF has a strong jerk in the initial stage. These issues in AOGL and VFPF will pose problems to the actuators, and thus, we do not discuss them further. NLGL and NSGL have comparable performances with NLGL performing marginally better. PLOS commands higher acceleration rates than NSGL to converge on the given path but has a smooth performance. In terms of cross-track errors, among NLGL, NSGL, and PLOS, NLGL has a similar performance to NSGL with NLGL having a larger overshoot as seen in Fig. 5(*c*). PLOS although faster is oscillatory in nature and also has a larger overshoot than NSGL. Qualitatively speaking, the settling time is better for NSGL as compared to PLOS and NLGL. This is in addition to the fact that NSGL has the least steady-state error (approximately 0.37 m) as against NLGL (approximately 0.57 m).

4.2 Two-Dimensional Circular Path. In this case, the UAV is commanded to follow a circle of radius 100 m with its center at

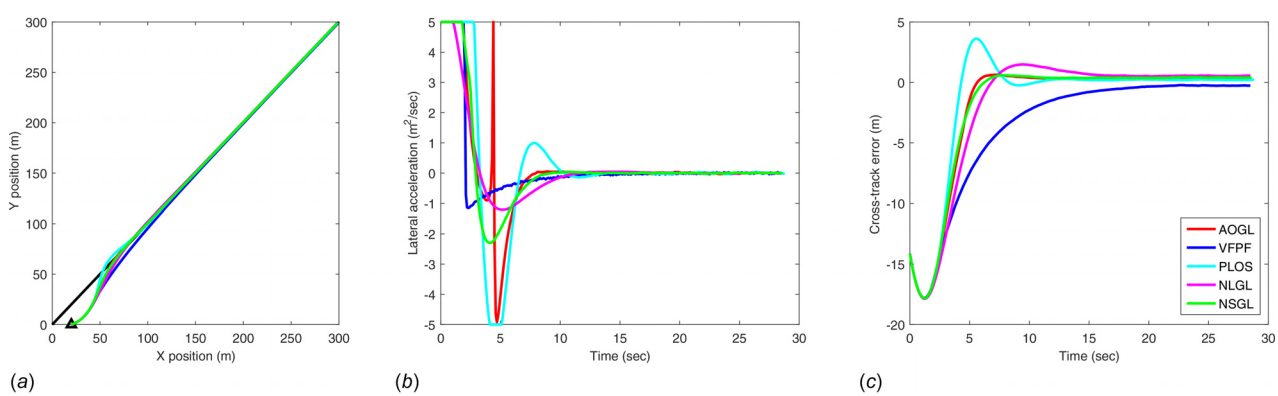

Fig. 5 Comparative plots demonstrating performance of various path following guidance laws on a straight line in 2D: (*a*) desired versus actual paths, (*b*) lateral accelerations, and (*c*) cross-track errors

Downloaded from https://asmedigitalcollection.asme.org/dynamicsystems/article-pdf/141/7/071008/6415992/ds_141_07_071008.pdf by guest on 05 July 2020

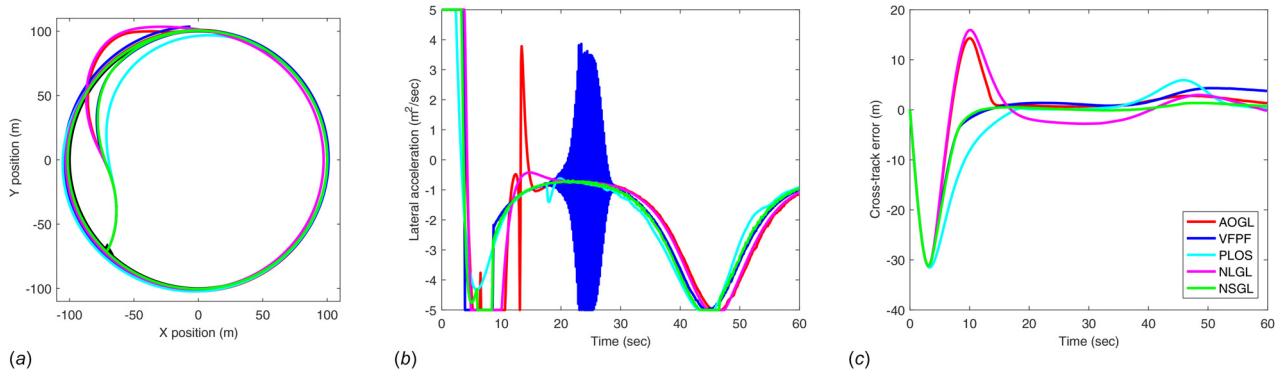

Fig. 6 Comparative plots demonstrating performance of various path following guidance laws on a circular path in 2D: (*a*) desired versus actual paths, (*b*) lateral accelerations, and (*c*) cross-track errors

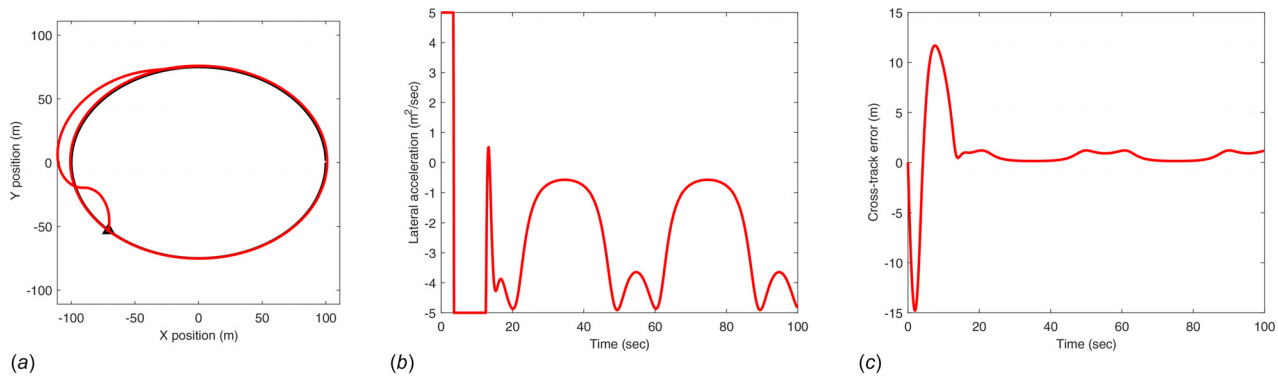

Fig. 7 Simulation results for NSGL on an elliptical path in 2D: (*a*) desired (black) versus actual (red) path, (*b*) lateral accelerations, and (*c*) cross-track errors

the origin (0,0)m. The simulation parameters for the UAV are retained from Sec. 4.1, except for the starting which is now at $(-70, -70)$m, and its initial velocity vector makes an angle of 45 deg with the X-axis. The gains are set to $k_1 = 2$, $k_2 = 4$, and $M_2 = 9$. The wind is now blowing with 4.5 ± 0.4 m/s in the direction toward the X-axis. The simulation results are presented in Fig. 6. As can be seen from Fig. 6(*b*), jerks are highest in AOGL with NLGL and NSGL having better performances. Also, VFPF exhibits chattering behavior in the presence of external disturbances (wind). All the guidance laws converge but the rates of convergence and steady-state errors vary. As can be seen from Fig. 6(*c*), NLGL and AOGL have large overshoots which leads to a delay in convergence. VFPF has an inferior steady-state performance with respect to others. NSGL and PLOS compare closely, but the settling time and the steady-state error of NSGL are relatively lesser. We can thus qualitatively say that as compared to the laws under consideration, NSGL has better performance.

4.3 Two-Dimensional Elliptic Path. Since the projection of an arbitrary circle in 3D onto a plane is an ellipse, we also evaluate the ability of the guidance law to track an ellipse in 2D case for the sake of completeness. An elliptical path centered at (0, 0) is chosen with the semimajor axis $a = 100$ m and the semiminor axis of $b = 75$ m. The initial position of UAV is on the ellipse at $(-70.7, -53.0)$m, and its initial vector is at the angle $\pi/4$ from the tangent on the ellipse. The simulation parameters for the UAV are again retained with a wind of 4.5 m/s that is blowing toward the positive X-axis. The gains are set to $k_1 = 0.5$, $k_2 = 4$, and $M_2 = 9$. The simulation results are presented in Fig. 7. The cross-track error plotted in Fig. 7(*c*) displays convergence with slight deviation from its converged value at the highest curvature which is stabilized periodically. These variations are within acceptable limits for all practical purposes.

4.4 Three-Dimensional Straight Line. The simulation results for the case of a 3D straight line path are presented in Fig. 8. The speed of the UAV is 15 m/s. The maximum lateral acceleration limits for u_h and u_v are taken to be 5 m/s^2 each. The initial position of the UAV is at (40, 10, 20)m. The initial azimuth and elevation angles of the velocity vector are 135 deg and 65 deg, respectively. The simulation parameters to calculate both the accelerations using the proposed guidance law remain the same as those chosen for the straight line case in Sec. 4.1. The UAV has to follow a path formed by joining W_1(0, 0, 0)m and W_2(300, 300, 300)m with a wind of 4.5 m/s acting as the external disturbance. The wind vector's azimuth angle is 135 deg and the elevation angle is 45 deg. The performance of the guidance law can be evaluated from the variation of lateral accelerations which is in the acceptable range, as can be observed in Fig. 8(*b*). The convergence of the cross-track errors eventually leading to the convergence of the UAV's track with its desired one in 3D can be observed in Fig. 8(*c*).

4.5 Three-Dimensional Circular Path. The simulation results for the case of a circular path in 3D are presented in Fig. 9. A circle of radius 150 m with its center located at (10, 20, 30)m is considered. The normal vector to the plane containing the circle has azimuth and elevation angles as 60 deg and 45 deg, respectively. The initial position of the UAV is set to $(-129, 13, 112)$m. The initial azimuth and elevation angles of the velocity vector are 135 deg and 60 deg, respectively. The speed of the UAV and the limits for maximum lateral acceleration inputs are kept unchanged from Sec. 4.4. The gains to calculate both the accelerations using

Downloaded from https://asmedigitalcollection.asme.org/dynamicsystems/article-pdf/141/7/071008/6415992/ds_141_07_071008.pdf by guest on 05 July 2020

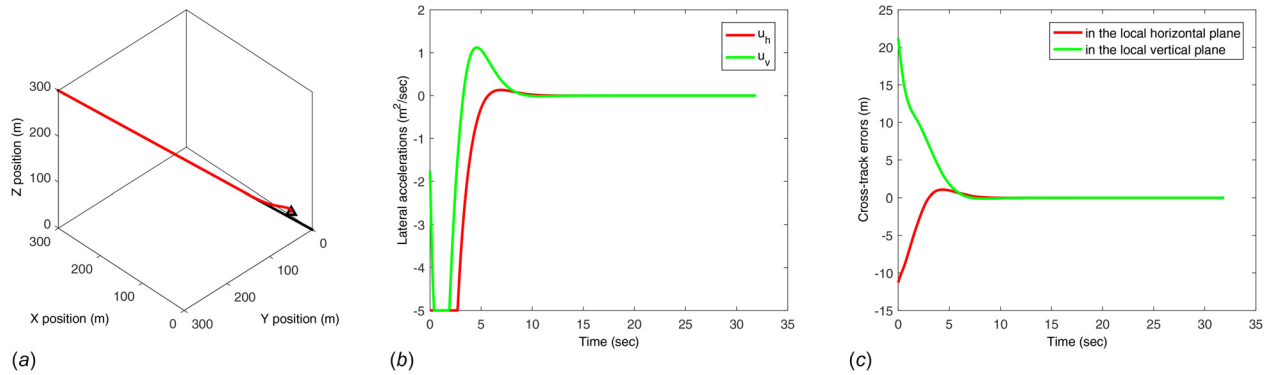

Fig. 8 Simulation results for NSGL on a straight line in 3D: (*a*) desired (black) versus actual (red) path, (*b*) lateral accelerations, and (*c*) cross-track errors

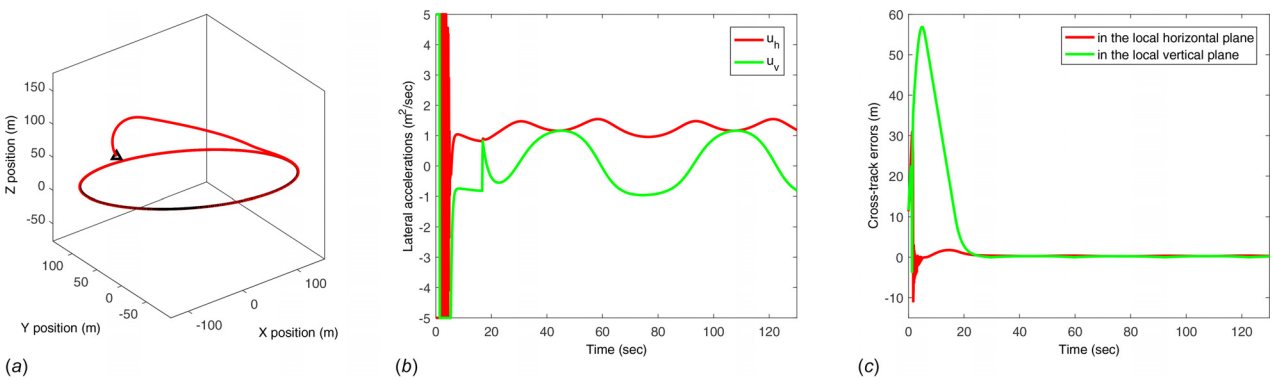

Fig. 9 Simulation results for NSGL on a circular path in 3D: (*a*) desired (black) versus actual (red) path, (*b*) lateral accelerations, and (*c*) cross-track errors

Fig. 10 X-UAV's Talon: The fixed-wing UAV platform that is used to perform experimental validation of the proposed guidance law

the proposed guidance law also remain the same as those chosen for the circular path case in Sec. 4.2. The rest of the assumptions, with respect to wind parameters, remain the same as in Sec. 4.4. The history lateral accelerations obtained from the simulations are plotted in Fig. 9(*b*), and cross-track errors are plotted in Fig. 9(*c*). The proposed guidance law can be observed to perform extremely well and converge to the desired path by limiting the lateral accelerations to their prescribed values.

5 Flight Test Results

This section details the hardware and autopilot setup used to implement the guidance law in real flight. Flight test results for 2D and 3D path following are presented thereafter.

5.1 Hardware. The proposed guidance law is implemented on one of the popular commercial UAV platforms Talon from X-UAV company [37], as shown in Fig. 10. The wingspan is 1.718 m and the wing area is 0.06 m^2. The maximum take-off weight of this airframe is 3.8 kg. The platform uses NTM Prop-Drive 35-48 1100 kv motor in conjunction with 11×5.5 in propeller mounted in pusher configuration. A 16,000 mAh Lipo 4-cell 20 °C battery gives an estimated cruise time of more than 40 min with a nominal airspeed of around 25 m/s.

5.2 Autopilot. The flight control unit is the 3DR PixhawkV2 controller board running the PX4 codebase [38]. The autopilot communicates with the ground station using a 3DR 915Mhz radio which is running the QGroundControl (ground control system) [39]. The autopilot has an extended Kalman filter for the state estimation which uses the global positioning system signal, the onboard inertial measurement unit, differential and absolute pressure sensors, and the onboard magnetometer to calculate the position, orientation, heading, and flight path angle of the vehicle. The codebase was modified to accommodate the proposed guidance law. A user can send commands such as desired velocity, altitude and path to the UAV while it is flying. Once a desired path is communicated, the nested saturation law is used to calculate the required lateral accelerations. The value of $u_{h_{\max}}$ is calculated from the lateral acceleration value associated with a maximum bank angle ($|\phi| \leq 40$ deg) using $u_{h_{\max}} = g \tan \phi_{\max}$, where g is the

Downloaded from https://asmedigitalcollection.asme.org/dynamicsystems/article-pdf/141/7/071008/6415992/ds_141_07_071008.pdf by guest on 05 July 2020

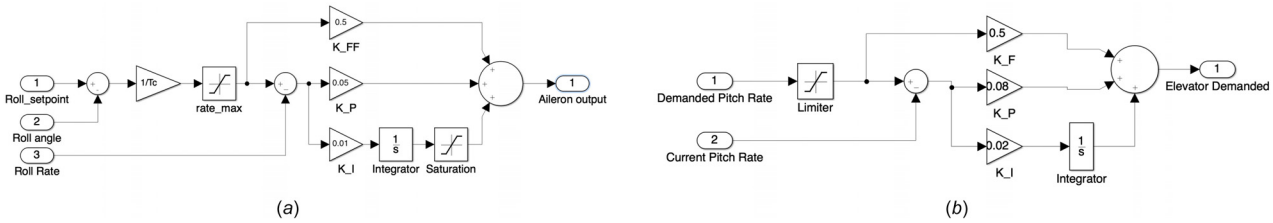

Fig. 11 Inner loop controllers that are used for flight tests: (*a*) roll and (*b*) pitch

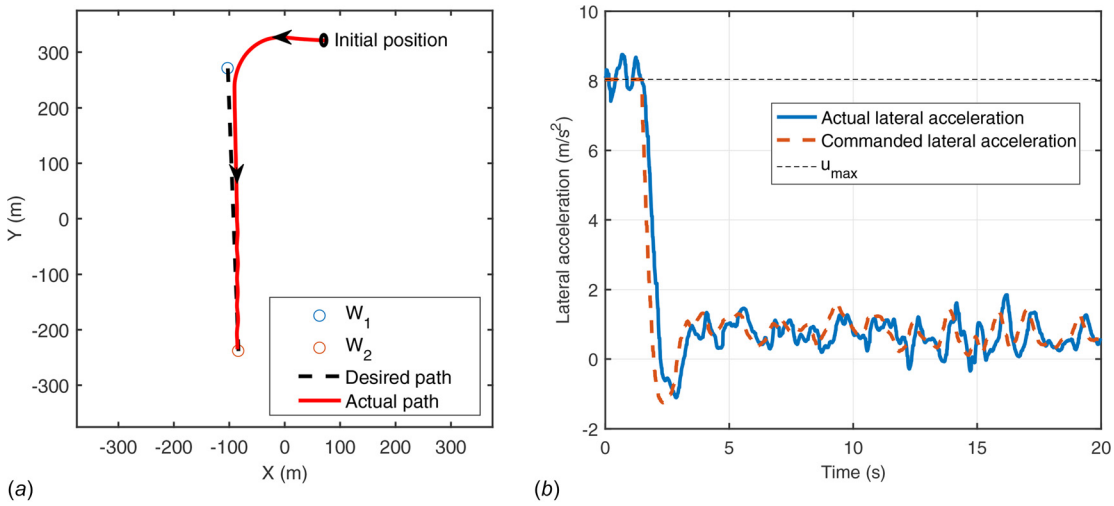

Fig. 12 Flight test results for a 2D straight line: (*a*) desired versus actual path and (*b*) commanded versus actual lateral acceleration

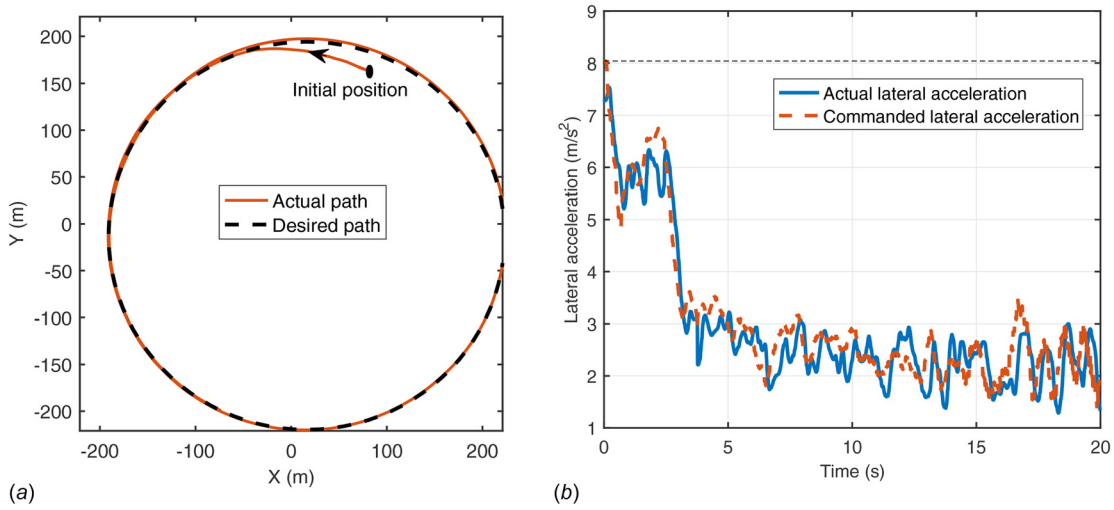

Fig. 13 Flight test results for a 2D circular path: (*a*) desired versus actual path and (*b*) commanded versus actual lateral acceleration

acceleration due to gravity. The desired roll angle ϕ is calculated from the desired lateral acceleration u_h using the relation $\tan\phi = (u_h/g)$, which is given as the control input to an inner loop roll controller,[4] as shown in Fig. 11(*a*). A PI controller on the desired roll rate (calculated from the desired and the current roll angles) with a feed-forward term is used to generate the aileron control command in the normalized range $[-1,1]$. The aileron output is then converted to a pulse width modulation signal to be provided to the actuators [38].

[4]https://github.com/PX4/ecl

Table 1 Parameters for the flight tests in the case of 3D rectangle path following

Projection	Parameter	Value
Vertical Plane	M_1	7
	M_2	5
	k_1	0.4
	k_2	3
Horizontal Plane	M_2	6
	k_1	2.5
	k_2	1.5

Downloaded from https://asmedigitalcollection.asme.org/dynamicsystems/article-pdf/141/7/071008/6415992/ds_141_07_071008.pdf by guest on 05 July 2020

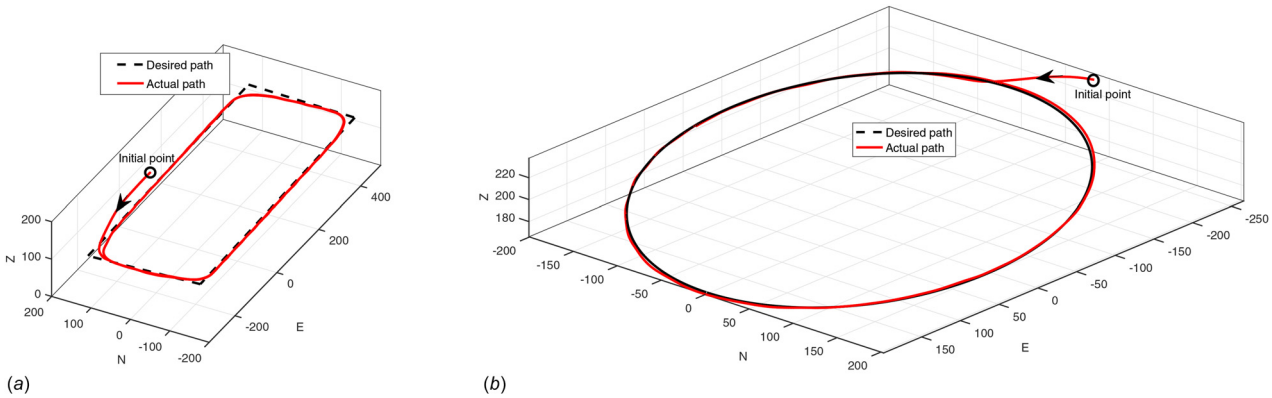

Fig. 14 Flight test results for 3D paths: (*a*) rectangular path and (*b*) circular path

The current PX4 architecture uses the total energy conservation system, which simultaneously controls the speed and the altitude using throttle to control total energy, and pitch to control the conversion of energy between potential and kinetic. In order to validate NSGL, which requires a constant speed, the total energy conservation system loop was tweaked to ignore its dependency on potential energy such that the throttle command maintains the desired constant airspeed. The pitch rate is commanded based on the desired vertical acceleration of UAV, as generated from NSGL, using the form $\dot{\theta}_d = (u_v/V)$ The desired $\dot{\theta}_d$ is given as input to a proportional-integral-derivative based inner loop that generates the required elevator commands.[4] The inner loop structure with the gains used is detailed in Fig. 11(*b*). The path inclinations and value of u_{v_m} were decided based on the maximum rate of climb achievable by the aircraft.

5.3 Two-Dimensional Paths. The wind speed is nearly zero. For the 2D cases, the controller is tested at a commanded altitude of 100 m and a commanded airspeed of 25 m/s. Initially, the aircraft is commanded to follow a straight line between two waypoints. The gains are tuned to $k_1 = 1.6$, $k_2 = 0.8$, and $M_2 = 3$. Figure 12(*a*) shows the desired path between two waypoints and the actual path followed by the UAV. Figure 12(*b*) shows the commanded versus achieved lateral acceleration for the first 20 s after the navigation command is communicated. It can be observed that the acceleration achieves its maximum value but drops as it starts converging, which is inline with the simulation results. The aircraft is then commanded to follow a circle of radius 150 m counter-clockwise around a fixed point in space. The gains are now tuned to $k_1 = 2$, $k_2 = 1$, and $M_2 = 4$. Figure 13(*a*) shows the desired path versus the actual path traveled by the UAV and Fig. 13(*b*) presents the commanded and achieved lateral accelerations, where it can be observed that the acceleration though achieves its maximum value initially, drops as it starts converging.

5.4 Three-Dimensional Paths. For the 3D flight tests, the UAV is initially commanded to follow a tilted rectangle whose two of the four adjoining vertices are at an altitude of 100 m and the rest two are at 150 m. Overshoot was avoided by setting a waypoint radius that changes the commanded waypoint when the aircraft is within a set radius of the current waypoint. The gains obtained from tuning are detailed in Table 1. The desired and actual paths are plotted in Fig. 14(*a*). The root-mean-square value of the distance error between the actual path and the nearest point on the desired path for one leg of the rectangle is found to be 1.9 m. The UAV is then commanded to follow a circular path with radius 200 m centered at (0,0) and tilted by 10 deg angle respect to the vertical. As discussed, the projection of a circle in 3D onto a plane is an ellipse. The equations (refer to Eq. (11*a*)) in Sec. 2.3 are used to obtain the errors. However, solving a fourth-order equation to get errors is an iterative process which can be difficult to implement given the hardware limitations. This is avoided by using an approximation that the closest point (x_p, y_p) is the point of intersection between the ellipse and a straight line joining the center of the ellipse and the UAV. As there can be two points of intersection, the closest point is obtained by comparing the distances. The approximation provides exact values when the UAV is on the ellipse, and when it is on the lines that are coincident with the axes. The results are presented in Fig. 14(*b*) and the gains are tabulated in Table 2. As can be seen, the aircraft is able to follow the path with a reasonable level of accuracy. The root-mean-square value of the distance error between the actual path and the nearest point on the desired path is found to be 2.46 m.

Table 2 Parameters for the flight tests in the case of 3D circular path following

Projection	Parameter	Value
Vertical Plane	M_1	7
	M_2	5
	k_1	0.2
	k_2	2.5
Horizontal Plane	M_2	4
	k_1	0.8
	k_2	2

6 Conclusion and Discussion

The paper presents a novel guidance law based on nested input saturation for straight line and circular paths in 3D space. The 3D path following problem is tackled by projecting the desired path onto two orthogonal planes and simultaneously solving the associated 2D path following problems. The stability of the guidance law, under certain conditions on the gain, is established for 2D cases using LaSalle's principle. Numerical simulations which also include wind disturbances are used to demonstrate its implementability in real-world applications. A comparison with the existing guidance laws is performed to establish its superior performance. The 2D and 3D cases of the proposed control law are successfully implemented on a fixed-wing UAV. Challenges while tuning the parameters in flight tests and testing in windy conditions will be taken up as a part of future work.

Nomenclature

d = cross-track error
u = commanded acceleration
v = inertial speed
$\mathbf{V}$ = inertial velocity vector

Downloaded from https://asmedigitalcollection.asme.org/dynamicsystems/article-pdf/141/7/071008/6415992/ds_141_07_071008.pdf by guest on 05 July 2020

γ = elevation angle
θ = pitch angle
ϕ = roll angle
χ = Azimuth angle
ψ = generalized heading angle

Subscripts

h = horizontal
v = vertical

Appendix: Proof for Theorem 3.1

The proof is divided into two sections for the sake of readability.

A.1 Separation Sets. Assume that M_2 and k_1 are selected such that $(M_2/k_1) < v_m$. The reason for this selection will be discussed in Sec. A.2. First, $\mathcal{S}$ can be divided into different sets based on the vector field of the closed-loop error trajectory. The vector field $\mathbf{F}$ is given by

$$\mathbf{F} = u\cos\tilde{\psi}\,\hat{\mathbf{i}} + \dot{d}\,\hat{\mathbf{j}} \tag{A1}$$

where $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ are the unit vectors along the X- and the Y-axis, respectively. In order to determine the direction $\ddot{d}$, consider the function

$$W_1 = \frac{1}{2}\dot{d}^2 \tag{A2}$$

and its time derivative can be obtained as

$$\dot{W}_1 = \dot{d}\,\ddot{d} = -\dot{d}\,\sigma_{M1}\left(\frac{k_1\dot{d} + \sigma_{M2}\big(k_2(k_1\dot{d} + d)\big)}{\cos\tilde{\psi}}\right)\cos\tilde{\psi} \tag{A3}$$

The direction associated with $\ddot{d} \in \mathcal{S}$ can be determined through $\dot{W}_1$. Whenever $\dot{W}_1$ is negative, $\ddot{d}$ is directed toward the d-axis, otherwise it is directed away from it. A group of sets, $\mathcal{S}_j \in \mathcal{S}$ for $j = 1, 2, \ldots, 6$, is found for which $\dot{W}_1 \le 0$ (by solving the inequality $\dot{W}_1 \le 0$), and is given below:

- $\mathcal{S}_1 = \{\dot{d}, d : 0 \le \dot{d} \le v_m \,\&\, k_2(k_1\dot{d} + d) \ge M_2\}$
- $\mathcal{S}_2 = \left\{\dot{d}, d : \frac{M_2}{k_1} \le \dot{d} \le V_{\max}\right\}$
- $\mathcal{S}_3 = \left\{\dot{d}, d : -v_m \le \dot{d} \le -\frac{M_2}{k_1}\right\}$
- $\mathcal{S}_4 = \{\dot{d}, d : -v_m \le \dot{d} \le 0 \,\&\, k_2(k_1\dot{d} + d) \le -M_2\}$
- $\mathcal{S}_5 = \{\dot{d}, d : 0 \le \dot{d} \le v_m \,\&\, |k_2(k_1\dot{d} + d)| \le M_2 \,\&\, (k_1 + k_1k_2)\dot{d} + k_2 d \ge 0\}$
- $\mathcal{S}_6 = \{\dot{d}, d : -v_m \le \dot{d} \le 0 \,\&\, |k_2(k_1\dot{d} + d)| < M_2 \,\&\, (k_1 + k_1k_2)\dot{d} + k_2 d < 0\}$

The union of the above sets $\mathcal{S}_1 \cup \mathcal{S}_2 \ldots \cup \mathcal{S}_6$ can be divided into $\mathcal{S}_o = \{\dot{d}, d : (M_2/k_1) \le |\dot{d}| \le v_m\}$ and $\mathcal{S}_i = \{\dot{d}, d : 0 \le |\dot{d}| \le (M_2/k_1)\}$. The convergence analyses for $\mathcal{S}_o$ and $\mathcal{S}_i$ are carried out separately. Figures 15 and 16 denote the trajectory evolution in $\mathcal{S}_o$ and $\mathcal{S}_i$, respectively, that take into account the sign of $\dot{W}$ in Eq. (A3). Note that the trajectory always starts in $\mathcal{S}$ because $|\dot{d}| \le v_m$. Let us also define the line $(k_1 + k_1k_2)\dot{d} + k_2 d = 0$ as L_1, the lines $k_2(k_1\dot{d} + d) = M_2$ as M_2 positive saturation lines, and $k_2(k_1\dot{d} + d) = -M_2$ as M_2 negative saturation lines. Let $((M_2/k_1), J_1)$ denote the point of intersection of M_2 saturation line with the line $\dot{d} = (M_2/k_1)$ and $(-(M_2/k_1), J_2)$ denote the point of intersection of M_2 negative saturation line with the line $\dot{d} = -(M_2/k_1)$. Similarly, let

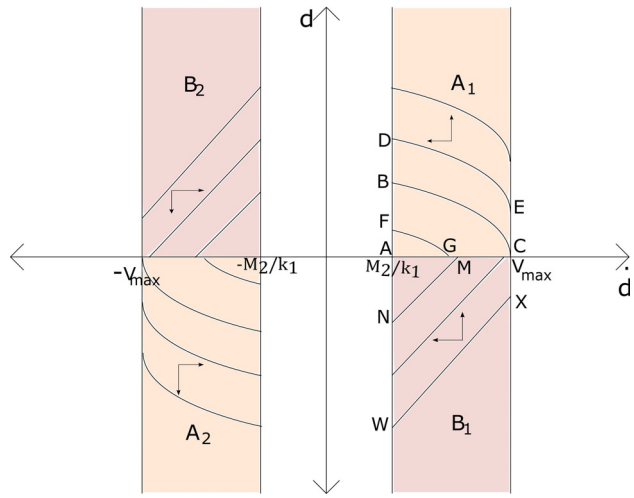

Fig. 15 Separation region, vector field F, and contours of Lyapunov function for set $\mathcal{S}_o$

$((M_2/k_1), P_1)$ be the point of intersection of L_1 with $\dot{d} = (M_2/k_1)$, and $(-(M_2/k_1), P_2)$ be the point of intersection of L_1 with $\dot{d} = -(M_2/k_1)$. It can be shown that $J_1 > P_1$ through proof by contradiction as follows. Let us assume $P_1 \ge J_1$, and J_1 can be obtained by solving $\dot{d} = (M_2/k_1)$ and M_2 saturation line equation simultaneously which gives $J_1 = (M_2(1 - k_1)/k_2)$. Similarly, the K_1 point can be obtained by solving $\dot{d} = (M_2/k_1)$ and L_1 simultaneously which gives $P_1 = (-M_2(1 + k_1)/k_2)$. Now, if $P_1 \ge J_1$

$$\begin{aligned} &\Rightarrow \frac{-M_2(1 + k_2)}{k_2} \ge \frac{M_2(1 - k_1)}{k_2}, \\ &\Rightarrow -1 - k_2 \ge 1 - k_2, \\ &\Rightarrow -1 \ge 1 \end{aligned} \tag{A4}$$

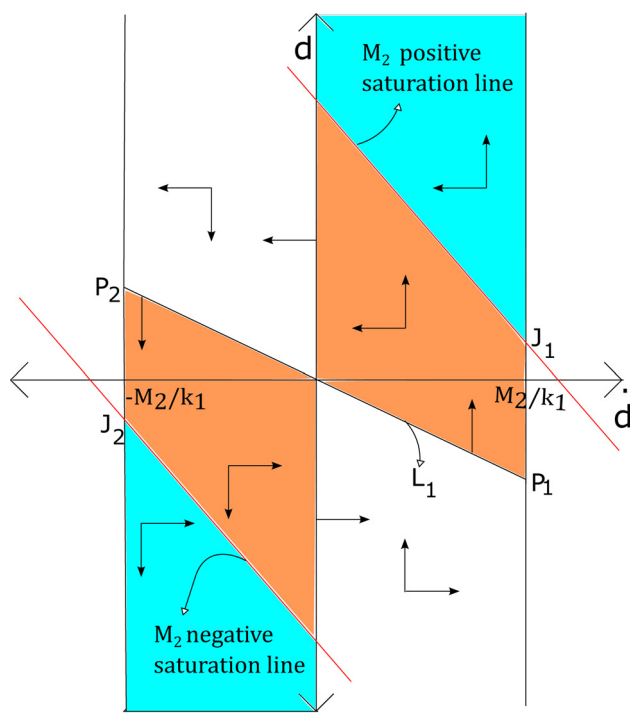

Fig. 16 Different separation sets based on the vector field F in $\mathcal{S}_i$; the arrows in the figure show the rate of change in d and $\dot{d}$

Downloaded from https://asmedigitalcollection.asme.org/dynamicsystems/article-pdf/141/7/071008/6415992/ds_141_07_071008.pdf by guest on 05 July 2020

Thus, the initial assumption that $P_1 \geq J_1$ is incorrect. A similar argument holds for J_2 and P_2. This analysis will be useful in subsequent discussion.

A.2 Lyapunov Analysis. In Sec. A.1, we have defined the vector field and divided the domain $\mathcal{S}$ based on the directions of vector field. It can be noted that the vector field does not always point toward the stationary point. In order to show the convergence, a geometric approach is used to construct Lyapunov functions in the domain $\mathcal{S}$ and then show that the area of the closed polygon reduces to zero as time evolves.

A.2.1 Convergence in the Outside Region. First, it is shown that any given trajectory in $\mathcal{S}_o$ will eventually enter the set $\mathcal{S}_i$. The objective is to find closed contours such that the trajectory moves from a contour of higher value to that of a lower value. To this end, the maximum inclinations of the vector field in different regions are found. Consider the set $\mathcal{A}_1 \triangleq \left\{d, \dot{d} : d \geq 0 \,\&\, \frac{M_2}{k_1} \leq \dot{d} \leq v_m\right\}$. The slope of the vector field ($\tan\delta$) at each point in phase plane with respect to the $-\dot{d}$-axis is given by

$$\tan\delta \triangleq \frac{\dot{d}}{-\ddot{d}} = \frac{\dot{d}}{\sigma_{M1}\left(\dfrac{k_1\dot{d} + \sigma_{M2}\left(k_2\left(k_1\dot{d} + d\right)\right)}{\cos\tilde{\psi}}\right)\cos\tilde{\psi}}$$

where $\cos\tilde{\psi} = \pm\sqrt{1 - (\dot{d}^2/V^2)}$. Note that $\pm$ will not affect the stability analysis, and the sign of $\cos\tilde{\psi}$ in denominator within the saturation function will cancel off (in sign) the $\cos\tilde{\psi}$ in the denominator of $\tan\delta$. Thus, the following stability analysis is performed considering the $+$ sign. For a constant $\dot{d}$, as d is increased, the slope remains nonincreasing. Thus, for a fixed $\dot{d}$, the maximum slope occurs along $d = 0$ (for $\mathcal{A}_1$). Now, it is important to find the point along $d = 0$ where the maximum slope occurs. Thus, by setting $d = 0$, it can be written as

$$\tan\delta^{\max} = \frac{\dot{d}}{\sigma_{M1}\left(\dfrac{k_1\dot{d} + \sigma_{M2}\left(k_2k_1\dot{d}\right)}{\cos\tilde{\psi}}\right)\sqrt{1 - \dfrac{\dot{d}^2}{V^2}}}$$

From the above equation, it is obvious that the maximum slope (∞) for $d = 0$ occurs at (v_m, 0), and then it is decreasing up until $((M_2/k_1), 0)$ which is a finite value. Refer to Fig. 15, the initial contour is created starting at the point $C((M_2/k_1), 0)$ and following a curve whose tangent slope is given by $\tan\delta$ evaluated at the corresponding $(\dot{d}, 0)$. Note that it will result in a continuous curve. The contour is completed by joining with point A. The contour denoted by ABCA can be slid up and down to create other contours. Note that the contours created by sliding ABC upward represent another contour represented by ADECA. While, sliding the contour downward results in AFGA after required truncation. The sliding operation ensures that the trajectory still points inward for every contour. The Lyapunov function at each point can be denoted as the area of the largest contour corresponding to that point. The Lyapunov function in the region is defined as the area of the largest contour containing the point. Note that by this construction, the dot product between the trajectory and the normal to the contour of Lyapunov function at every point is less than or equal 0. Since $\mathcal{A}_2 \triangleq \{d, \dot{d} : d \leq 0 \,\&\, -v_m \leq \dot{d} \leq (M_2/k_1)\}$ is antisymmetric to $\mathcal{A}_1$, similar contours can be drawn in $\mathcal{A}_2$.

Now, consider the region $\mathcal{B}_1 \triangleq \{d, \dot{d} : d \leq 0 \,\&\, (M_2/k_1) \leq \dot{d} \leq v_m\}$. Constructing contours for this region is fairly simple. Refer to Fig. 15, one of the edges can be a straight line with any positive slope. One of the contours is represented by AMNA and another by ACXWA. The Lyapunov function at a point in this region is again the area of the largest polygon containing the point. Note that by this construction, the dot product between the trajectory and the normal to the contour of Lyapunov function at every point is less than 0. Thus, trajectory always moves from higher to lower valued contour. Similar arguments follow for region $\mathcal{B}_2 \triangleq \{d, \dot{d} : d \geq 0 \,\&\, -v_m \leq \dot{d} \leq -(M_2/k_1)\}$.

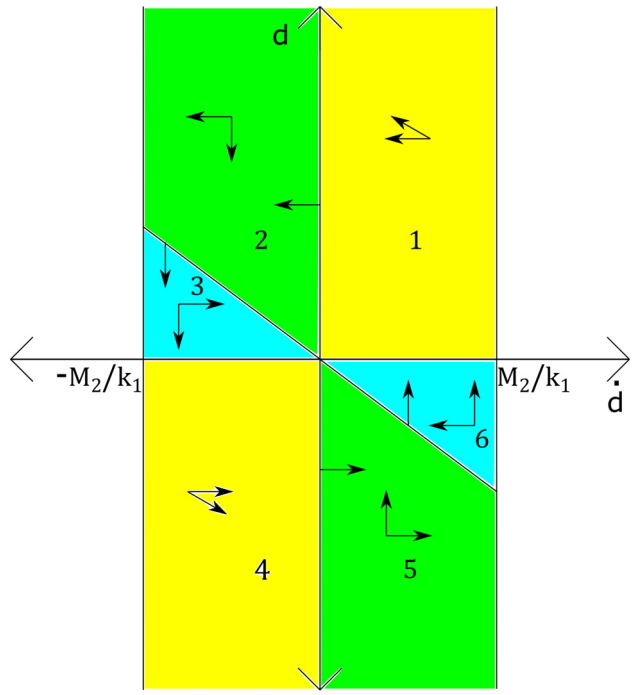

Fig. 17 Modified separation regions with different possible directions of evolution of the error trajectory

The Lyapunov functions in set $\mathcal{S}_o$ is represented by a set of nested closed curves or polygons. It has been proved that the trajectory either moves from a contour of higher value to lower value or remains on the same contour. Thus, it can be observed that if the trajectory starts in $\mathcal{B}_1$, then it either enters $\mathcal{S}_i$ or $\mathcal{A}_1$. Similarly, if the trajectory originates in $\mathcal{B}_2$, then it either enters $\mathcal{S}_i$ or $\mathcal{A}_2$. Once the trajectory is in $\mathcal{A}_1$ or $\mathcal{A}_2$, it eventually enters the set $\mathcal{S}_i$. Thus, the above arguments result in the trajectory eventually entering the region $\mathcal{S}_i$.

A.2.2 Convergence in the Inner Region. As proven earlier, the trajectory enters into the set $\mathcal{S}_i$. Now, the conditions on the gains can be obtained by choosing an appropriate Lyapunov function to prove convergence. In the process of choosing a positive definite Lyapunov function, $\mathcal{S}_i$ is divided into six regions, which are shown in Fig. 17. The objective is to construct a closed polygon in $\mathcal{S}_i$ such that $\mathbf{F}$ points inward on its contour. It requires finding a set of nested closed polygons whose hypersurfaces are oriented in $\mathcal{S}_i$ such that $\mathbf{a}_i.\mathbf{F} \leq 0$, where $\mathbf{a}_i$ is normal to the ith hypersurface pointing away from the origin.

Consider region 1 in Fig. 17. For an arbitrary orientation of the hypersurface, the condition $\mathbf{a}_i.\mathbf{F} \leq 0$ may not be satisfied. It is required to find θ (where θ is the inclination of the hypersurface of the polygon in region 1 measured with respect to the $-\dot{d}$ axis) such that $\mathbf{a}_i.\mathbf{F} \leq 0$, i.e., the vector field always points inward. Subsequently, the maximum inclination of the vector field in region 1 has to be found. The maximum slope of the vector field ($\tan\delta$) at each point in phase plane with respect to the $-\dot{d}$-axis (in region 1 and on $d = 0$ axis) is given by

$$\tan\delta = \frac{\dot{d}}{\sigma_{M1}\left(\dfrac{k_1\dot{d} + \sigma_{M2}\left(k_2k_1\dot{d}\right)}{\cos\tilde{\psi}}\right)\cos\tilde{\psi}} \tag{A5}$$

Downloaded from https://asmedigitalcollection.asme.org/dynamicsystems/article-pdf/141/7/071008/6415992/ds_141_07_071008.pdf by guest on 05 July 2020

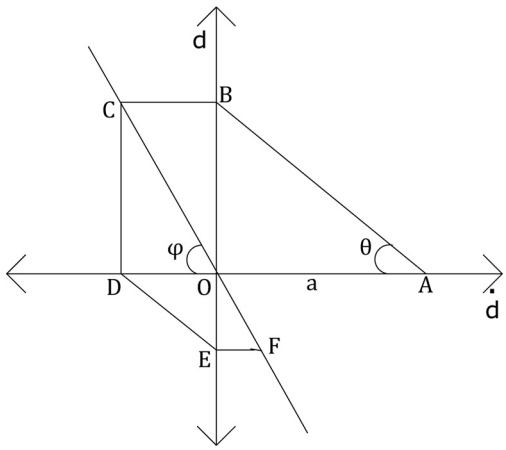

Fig. 18 Conditions on θ and ϕ to obtain a closed polygon

Plugging in $\dot{d} = (M_2/k_1)$ into the previous equation, it can be written as

$$\tan\delta_{\max} = \frac{\dfrac{M_2}{k_1}}{\sigma_{M1}\left(\dfrac{M_2 + \sigma_{M2}(k_1 M_2)}{\sqrt{1 - \dfrac{M_2{}^2}{v_m^2 k_1{}^2}}}\right)\sqrt{1 - \dfrac{M_2{}^2}{v_m^2 k_1{}^2}}} \tag{A6}$$

Since $|M_2/k_1| < v_m$ (the assumption in Sec. A.1), the above equation is valid. This condition ensures that the maximum slope in region 1 is bounded. Thus, if the vector field points inward at the contour of the polygon, the condition $\tan\theta \geq \tan\delta_{\max}$ (refer to Fig. 18) must hold. Thus, the inclination of the hypersurface (θ) in region 1 is considered as $\delta_{\max}$. Note that region 4 is antisymmetric to region 1; hence, the maximum slope in region 4 is the same as the one in region 1.

The phase plane is shown in Fig. 17. It can be observed that in regions 2 and 5, if we take horizon lines as edges of a polygon, the vector field will point inward. Similarly, when the vertical lines are considered to be the edges of the polygons in regions 3 and 6, the vector field will point inward. However, there exists difficulty in determining an edge in region 6 to close a polygon. It is because the slope of the vector field at the edge may point outward. To tackle this issue, some trigonometric relations can be employed. Let us consider a contour as shown in Fig. 18, and let the length of edge the OA be a. To ensure that the slope of **F** points inward, it follows that $EF \leq OA$. The length of the edge EF can be determined as $a\tan^2\theta\cot^2\phi$, and as per the requirement, $EF < OA$

$$a\tan\theta^2\cot\phi^2 < a$$

$$\Rightarrow \frac{\dfrac{M_2}{k_1}}{\sigma_{M1}\left(\dfrac{M_2 + \sigma_{M2}(k_1 M_2)}{\sqrt{1 - \dfrac{M_2{}^2}{v_m^2 k_1{}^2}}}\right)\sqrt{1 - \dfrac{M_2{}^2}{v_m^2 k_1{}^2}}} < \frac{k_1 + k_1 k_2}{k_2}$$

Note that the above contour analysis was possible because $J_1 > P_1$ as proved in Sec. 6. This allows the angle ϕ in this region that can be determined only by L_1.

Thus, the conditions on gains are given by

$$(1)\ \frac{M_2}{k_1} < v_m$$

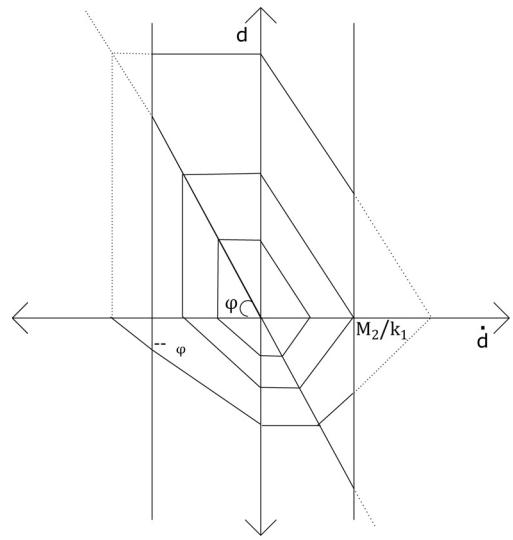

Fig. 19 Construction of Lyapunov function using nested polygons

$$(2)\ \frac{\dfrac{M_2}{k_1}}{\sigma_{M1}\left(\dfrac{M_2 + \sigma_{M2}(k_1 M_2)}{\sqrt{1 - \dfrac{M_2{}^2}{v_m^2 k_1{}^2}}}\right)\sqrt{1 - \dfrac{M_2{}^2}{v_m^2 k_1{}^2}}} < \frac{k_1 + k_1 k_2}{k_2}$$

With the above conditions, a Lyapunov function is built as a set of nested polygons as shown in Fig. 19. Note that the Lyapunov function at a point can be defined as the area of the largest polygon containing that point. For the proposed path following guidance law, it is not required that an initial condition falls in the set $\mathcal{S}_i$. As argued earlier, the error trajectory will enter $\mathcal{S}_i$. Once it is in $\mathcal{S}_i$, the error trajectory will pass from larger to smaller polygons, and thus, the area will decrease continuously. The error trajectories at vertices may point along the edges. Therefore, the time derivative of Lyapunov function is negative semidefinite. The stationary point (0, 0) is unique and is the largest invariant set among the separation regions. Thus, by LaSalle's theorem [40], every solution will be asymptotically stable. Therefore, if k_1, k_2, and M_2 are chosen as given above, then the convergence of the error trajectory to zero is guaranteed.

References

[1] Kanistras, K., Martins, G., Rutherford, M. J., and Valavanis, K. P., 2013, "A Survey of Unmanned Aerial Vehicles (UAVs) for Traffic Monitoring," International Conference on Unmanned Aircraft Systems (ICUAS), Atlanta, GA, May 28–31, pp. 221–234.

[2] Turner, I. L., Harley, M. D., and Drummond, C. D., 2016, "UAVs for Coastal Surveying," Coastal Eng., **114**, pp. 19–24.

[3] Costa, F. G., Ueyama, J., Braun, T., Pessin, G., Osório, F. S., and Vargas, P. A., 2012, "The Use of Unmanned Aerial Vehicles and Wireless Sensor Network in Agricultural Applications," IEEE International Geoscience and Remote Sensing Symposium (IGARSS), Munich, Germany, July 22–27, pp. 5045–5048.

[4] Maza, I., Caballero, F., Capitán, J., Martínez-de Dios, J. R., and Ollero, A., 2011, "Experimental Results in Multi-UAV Coordination for Disaster Management and Civil Security Applications," J. Intell. Rob. Syst., **61**(1), pp. 563–585.

[5] Mathew, N., Smith, S. L., and Waslander, S. L., 2015, "Planning Paths for Package Delivery in Heterogeneous Multirobot Teams," IEEE Trans. Autom. Sci. Eng., **12**(4), pp. 1298–1308.

[6] Yoshitani, N., 2010, "Flight Trajectory Control Based on Required Acceleration for Fixed-Wing Aircraft," 27th International Congress of the Aeronautical Sciences, Nice, France, Sept. 19–24.

[7] Rhee, I., Park, S., and Ryoo, C., 2010, "A Tight Path Following Algorithm of an Uas Based on PID Control," SICE Annual Conference 2010, Taipei, Taiwan, Aug. 18–21, pp. 1270–1273.

[8] Ratnoo, A., Sujit, P., and Kothari, M., 2011, "Adaptive Optimal Path Following for High Wind Flights," IFAC Proc. Vol., **44**(1), pp. 12985–12990.

[9] Cho, N., Kim, Y., and Park, S., 2015, "Three-Dimensional Nonlinear Differential Geometric Path-Following Guidance Law," J. Guid., Control, Dyn., **38**(12), pp. 2366–2385.

Downloaded from https://asmedigitalcollection.asme.org/dynamicsystems/article-pdf/141/7/071008/6415992/ds_141_07_071008.pdf by guest on 05 July 2020

[10] Cunha, R., Antunes, D., Gomes, P., and Silvestre, C., 2006, "A Path-Following Preview Controller for Autonomous Air Vehicles," AIAA Paper No. 2006-6715.
[11] Yamasaki, T., Balakrishnan, S. N., and Takano, H., 2012, "Integrated Guidance and Autopilot for a Path-Following UAV Via High-Order Sliding Modes," American Control Conference (ACC) , Montreal, QC, Canada, June 27–29, pp. 143–148.
[12] Liu, C., McAree, O., and Chen, W.-H., 2012, "Path-Following Control for Small Fixed-Wing Unmanned Aerial Vehicles Under Wind Disturbances," Int. J. Robust Nonlinear Control, **23**(15), pp. 1682–1698.
[13] Jung, D., and Tsiotras, P., 2008, "Bank-to-Turn Control for a Small Uav Using Backstepping and Parameter Adaptation," IFAC Proc. Vol., **41**(2), pp. 4406–4411.
[14] Beard, R. W., Ferrin, J., and Humpherys, J., 2014, "Fixed Wing UAV Path Following in Wind With Input Constraints," IEEE Trans. Control Syst. Technol., **22**(6), pp. 2103–2117.
[15] Nelson, D. R., Barber, D. B., McLain, T. W., and Beard, R. W., 2007, "Vector Field Path Following for Miniature Air Vehicles," IEEE Trans. Rob., **23**(3), pp. 519–529.
[16] Lawrence, D. A., Frew, E. W., and Pisano, W. J., 2008, "Lyapunov Vector Fields for Autonomous Unmanned Aircraft Flight Control," J. Guid., Control, Dyn., **31**(5), pp. 1220–1229.
[17] Yamasaki, T., Takano, H., and Baba, Y., 2009, "Robust Path-Following for UAV Using Pure Pursuit Guidance," *Aerial Vehicles*, InTech, London.
[18] Morales, J., Martínez, J. L., Martínez, M. A., and Mandow, A., 2009, "Pure-Pursuit Reactive Path Tracking for Nonholonomic Mobile Robots With a 2D Laser Scanner," EURASIP J. Adv. Signal Process., **2009**, p. 935237.
[19] Osborne, J., and Rysdyk, R., 2005, "Waypoint Guidance for Small UAVs in Wind," AIAA Paper No. 2005-6951.
[20] Rysdyk, R., 2006, "Unmanned Aerial Vehicle Path Following for Target Observation in Wind," J. Guid. Control Dyn., **29**(5), pp. 1092–1100.
[21] Ramana, M., Varma, S. A., and Kothari, M., 2016, "Motion Planning for a Fixed-Wing UAV in Urban Environments," IFAC-PapersOnLine, **49**(1), pp. 419–424.
[22] Curry, R., Lizarraga, M., Mairs, B., and Elkaim, G. H., 2013, "L_2^+, an Improved Line of Sight Guidance Law for UAVs," American Control Conference (ACC), Washington, DC, June 17–19, pp. 1–6.
[23] Wit, J. S., 2000, "Vector Pursuit Path Tracking for Autonomous Ground Vehicles," Ph.D. thesis, University of Florida, Gainesville, FL.
[24] Park, S., Deyst, J., and How, J. P., 2007, "Performance and Lyapunov Stability of a Nonlinear Path-Following Guidance Method," J. Guid., Control, Dyn., **30**(6), pp. 1718–1728.
[25] Breivik, M., and Fossen, T. I., 2005, "Principles of Guidance-Based Path Following in 2D and 3D," 44th IEEE Conference on Decision and Control (CDC), Seville, Spain, Dec. 12–15, pp. 627–634.
[26] Shanmugavel, M., Tsourdos, A., White, B., and Żbikowski, R., 2010, "Co-Operative Path Planning of Multiple UAVs Using Dubins Paths With Clothoid Arcs," Control Eng. Pract., **18**(9), pp. 1084–1092.
[27] Johnson, E., Calise, A., and Corban, J. E., 2002, "A Six Degree-of-Freedom Adaptive Flight Control Architecture for Trajectory Following," AIAA Paper No. 2002-4776.
[28] Murray, R. M., 1996, "Trajectory Generation for a Towed Cable System Using Differential Flatness," IFAC Proc. Vol., pp. 395–400.
[29] Rathinam, M., and Murray, R. M., 1998, "Configuration Flatness of Lagrangian Systems Underactuated by One Control," SIAM J. Control Optim., **36**(1), pp. 164–179.
[30] Sujit, P. B., Saripalli, S., and Sousa, J. B., 2014, "Unmanned Aerial Vehicle Path Following: A Survey and Analysis of Algorithms for Fixed-Wing Unmanned Aerial Vehicles," IEEE Control Syst. Mag., **34**(1), pp. 42–59.
[31] Pelizer, G. V., da Silva, N. B. F., and Branco, K. R. L. J., 2017, "Comparison of 3D Path-Following Algorithms for Unmanned Aerial Vehicles," International Conference on Unmanned Aircraft Systems (ICUAS), Miami, FL, June 13–16, pp. 498–505.
[32] Zhao, S., Wang, X., Zhang, D., and Shen, L., 2018, "Curved Path Following Control for Fixed-Wing Unmanned Aerial Vehicles With Control Constraint," J. Intell. Rob. Syst., **89**(1–2), pp. 107–119.
[33] Yamasaki, T., Balakrishnan, S., and Takano, H., 2013, "Separate-Channel Integrated Guidance and Autopilot for Automatic Path-Following," J. Guid., Control, Dyn., **36**(1), pp. 25–34.
[34] Teel, A. R., 1992, "Global Stabilization and Restricted Tracking for Multiple Integrators With Bounded Controls," Syst. Control Lett., **18**(3), pp. 165–171.
[35] Kothari, M., Postlethwaite, I., and Gu, D.-W., 2014, "Motion Planning for a Fixed-Wing UAV in Urban Environments," J. Intell. Rob. Syst., **74**(3), pp. 1013–1028.
[36] Pattanaik, A., and Kothari, M., 2016, "A Novel Guidance Law With Input Saturation," 55th Conference on Decision and Control (CDC), Las Vegas, NV, Dec. 12–14, pp. 4148–4153.
[37] Hugues, 2019, "X-UAV Talon build thread (planned with PIXHAWK autopilot)," accessed Apr. 21, 2014, https://diydrones.com/profiles/blogs/xuav-talon-build-thread-planned-with-pixhawk-autopilot
[38] Meier, L., Honegger, D., and Pollefeys, M., 2015, "PX4: A Node-Based Multi-threaded Open Source Robotics Framework for Deeply Embedded Platforms," IEEE International Conference on Robotics and Automation (ICRA), Seattle, WA, May 26–30.
[39] Zurich, E., 2013, "QGroundControl GCS," accessed Mar. 20, 2019, http://qgroundcontrol.com
[40] Khalil, H. K., 2002, *Nonlinear Systems*, Prentice Hall, Upper Saddle River, NJ, Chap. 9.

Downloaded from https://asmedigitalcollection.asme.org/dynamicsystems/article-pdf/141/7/071008/6415992/ds_141_07_071008.pdf by guest on 05 July 2020